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Brightlearn : Exercise 2

30 April 2026

1. List all distinct departments in the students table;

```
SELECT DISTINCT department  
FROM students;
```



department
IT
HR
Finance

2. Get the average age of students per department:

```
SELECT department,  
       AVG(age) AS avg-age  
FROM students  
GROUP BY department;
```

↘

department	avg-age
IT	20.5
HR	22
Finance	23

3. Show department with more than 1 student :

```
SELECT department, COUNT(*) AS student-count  
FROM students  
GROUP BY department  
HAVING COUNT(*) > 1;
```

↓

department	student-count
IT	2
HR	2

4. SELECT student-id,
name,
age,
department

FROM students
WHERE age BETWEEN 21 AND 23;

↓

Student-id	name	age	department
2	Bob	22	HR
3	charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

5. SELECT studentId,
 name,
 age,
 department
 FROM students
 WHERE age > 21 AND department IN ('IT', 'HR');



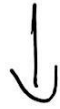
Student-Id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department,
 SUM(credits) AS total-credit
 FROM courses
 GROUP BY department
 HAVING total-credits > 5;



department	total-credit
IT	11

7. SELECT course-id,
department,
course-name,
credits
FROM courses
WHERE credits < 4;



course-id	department	credits
101	SQL Basics	3
104	Excel	2
105	Statistics	3

8. SELECT course-id, course-name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;



course-id	course-name	credits
102	Python	4
103	Data science	4
101	SQL Basics	2

9. SELECT MAX(grade) AS max-grade,
 MIN(grade) AS min-grade,
 AVG(grade) AS avg-grade
 FROM enrollments;
~~max-gr~~ ↓ outcome

max-grade	min-grade	avg-grade
90	78	84,6

10. SELECT course-id, COUNT(enrollment-id) AS enrollment-count
 FROM enrollments
 GROUP BY course-id;

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

11. SELECT department, SUM(salary) AS total-salary,
 SUM(bonus) AS total-bonus
 FROM salaries
 GROUP BY department;

department	total-salary	total-bonus
IT	122000	10500
HR	109000	7500
Finance	70000	600

12. SELECT department,
 AVG(salary) AS avg-salary
 FROM salaries
 GROUP BY department
 HAVING avg-salary > 55000;

department	avg-salary
IT	61000
Finance	70000

13. SELECT employee-id,
 name,
 salary,
 bonus,
 salary + bonus AS total-compensation
 FROM Salaries
 WHERE salary + bonus > 60000;



employee-id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. SELECT department
 SUM(budget) AS total-budget,
 AVG(budget) AS avg-budget
 FROM projects
 GROUP BY department
 HAVING AVG(budget) > 70000;

→

department	total-budget	avg-budget
IT	270000	135000
Finance	80000	80000

15. SELECT project-id,
 project-name,
 department,
 budget
 FROM Projects
 WHERE budget BETWEEN 80000 AND 120000
 AND department <> 'Marketing';



project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	80000